Binary Encoding
ASCII Encoding

CSE 101 Section 03

Introduction to Algorithms and Computing

Fall 2022

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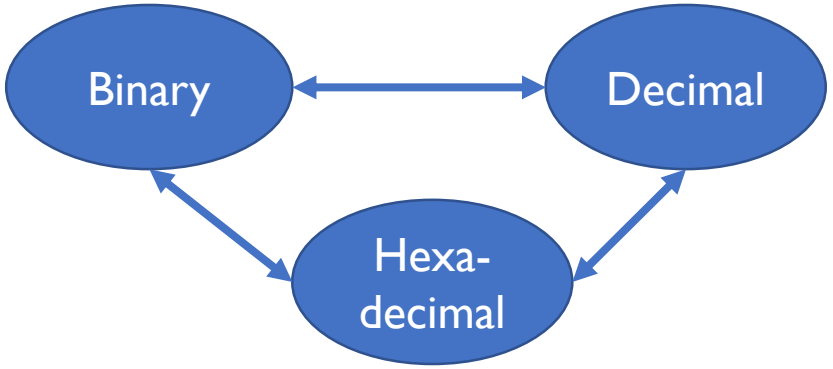
In the previous lecture

- We introduced different numbering systems:
  - Decimal
  - Binary
  - Hexadecimal

- We covered the approached to convert numbers from one system to another.



Why are we learning this?

Why does all of this matter?

- Humans think about numbers in base 10 but computers think about numbers in base 2
- How is it that computers can do all of the amazing things that they do?
  - Binary encoding

Numerical Encoding

FACT: You can represent anything countable using numbers!

- Need to agree on an encoding
- Kind of like learning a new language

Examples:

- Decimal Numbers: 0→0b0, 1→0b1, 2→0b10, etc.
- English Letters: BJC→0x424A43, yay→0x796179
- Emoticons: 😄 0x0, 😟 0x1, 😎 0x2, 😇 0x3, 😈 0x4, 👋 0x5

Binary Encoding

With N binary digits, how many things can you represent? 2^N

- Need N bits to represent *n* things, where $2^N \geq n$
- Example: 5 bits for alphabet because $2^5 = 32 > 26$

A binary digit is known as a bit

A group of 4 bits (1 hex digit) is called a nibble

A group of 8 bits (2 hex digits) is called a byte

bit → 2 things, nibble → 16 things, byte → 256 things

So what does it mean?

A sequence of bits can have many meanings!

Consider the hex sequence 0x4E6F21

Common interpretations include:

- The decimal number 5140257
- The characters "No!"
- The background colour of this slide
- The real number 7.203034×10^{-39} [floating point]

ASCII Encoding

Introduction

ASCII stands for *American Standard Code for Information Interchange*.

How is it related to binary encoding?

- Binary encoding is a general term used for a method of encoding anything countable (characters, decimal numbers, etc).

- ASCII is a globally accepted conventions of encoding *characters*, and was the most commonly used binary encoding scheme for more than three decades.

Why do we need ASCII?

- Computer programs are pretty good at displaying text on the screen.
- A good example of that is this text you are currently reading 😀

- But even though computers are good at displaying text on the screen, they can't actually store any text/characters in their memory, only numbers (or rather combinations of bits that represent different numbers).

- So how do they deal with this problem? Well, each type of character (a, b, c etc.) is <u>mapped</u> to a number (like 97, 98, 99 etc.)
- <u>ASCII encoding</u> is the convention that keep track of this mapping.

More about ASCII

- ASCII encoding uses exactly 1 bytes (8 bits) to encode characters.

- Based on above, how many characters can we encode with ASCII?

  $2^8 = 256$ characters

- Examples:

' '	32	00100000	'A'	65	01000001
'0'	48	00110000	'a'	97	01100001
'1'	49	00110001	'™'	128	10000000

ASCII table

- The publicly available ASCII table contains the mapping of 256 characters into their binary representations.

- Can be used to encode and decode any set of characters.

- Available on the course website, or using a Google search.

Next…

Two's Complement